

Bahram Behzadian

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Google Scholar | U.S. permanent resident

Reinforcement Learning | Post-Training | Evaluation

Profile

RL researcher (Ph.D.) working on policy improvement with imperfect evaluators and verifier feedback, including finite-depth error bounds for recurrent value evaluators (UPI-TRM) and robust MDPs under model uncertainty. At Meta, I build correctness and monitoring systems for large-scale online experimentation over ranking models, detecting stale, biased, or inconsistent signals. Interested in RL post-training for reasoning and long-horizon agents, including durable memory and reliable tool-use and search decisions under uncertainty.

Selected Research

UPI-TRM: Policy Improvement under Truncated Evaluators with Verifier-Style Rewards
ICLR 2026 Workshop on AI with Recursive Self-Improvement

Co-developed finite-depth error bounds for recurrent value evaluators used in policy improvement without assuming the recurrence converges. The analysis separates endpoint discrepancy from Bellman-residual error and links centered advantage error to conservative policy-improvement bounds. Evaluated the analysis on an exact finite MDP and a fixed set of Sudoku states.

Robust and Risk-Sensitive Markov Decision Processes
NeurIPS 2021 and AISTATS 2021

Developed scalable algorithms for L_∞ -constrained S-rectangular robust MDPs and percentile criteria under model uncertainty.

Experience

2022–Present Research Scientist, *Meta*

- Work on a Python reference implementation of ads experiment metrics for independent cross-checks of SQL-based correctness calculations.
- Develop rule-based data-quality monitoring and statistical diagnostics for stale, biased, or inconsistent signals.
- Turn production failure analyses into automated checks for overlapping data slices and attribution double counting in online experimentation over ranking models.

2018 AI Engineer Intern, *Envio, Inc.*, Dover, NH

Developed learning-based methods for vehicle routing and scheduling under operational constraints and uncertain execution.

Selected Publications

Bahram Behzadian, Brett Daley, Gopeshh Subbaraj, Houssam Nassif
UPI-TRM: Policy Improvement under Truncated Evaluators with Verifier-Style Rewards
ICLR 2026 Workshop on AI with Recursive Self-Improvement [Paper]

Bahram Behzadian, Marek Petrik, Chin Pang Ho
Fast Algorithms for L_∞ -Constrained S-Rectangular Robust MDPs
Conference on Neural Information Processing Systems (NeurIPS), 2021 [Paper]

Bahram Behzadian, Reazul Hasan Russel, Marek Petrik, Chin Pang Ho
Optimizing Percentile Criterion using Robust MDPs
International Conference on Artificial Intelligence and Statistics (AISTATS), 2021 [Paper]

Bahram Behzadian, Soheil Gharatappeh, Marek Petrik
Fast Feature Selection for Linear Value Function Approximation
International Conference on Automated Planning and Scheduling (ICAPS), 2019 [Paper]

Bahram Behzadian, Pratik Agarwal, Wolfram Burgard, Gian Diego Tipaldi
Monte Carlo Localization in Hand-Drawn Maps
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2015 [Paper]

Complete publication list: [Publications page](#) | [Google Scholar](#)

Technical Skills

Programming: Python, PyTorch.

Evaluation and experimentation: experiment-metric correctness, statistical diagnostics, anomaly and data-quality monitoring, ML auditing.

RL and post-training: robust MDPs, policy improvement, verifier-style rewards and RL with verifiable rewards; hands-on PPO and DPO experiments on small language models for arithmetic; working knowledge of GRPO and reward modeling.

Education

- 2015–2022 Ph.D. in Computer Science, *University of New Hampshire*
Dissertation: *Efficient Data-Driven Robust Policies for Reinforcement Learning* (advisor: Marek Petrik).
M.S. in Computer Science awarded en route (2018). Thesis: *Representation Learning for Reinforcement Learning via Singular Value Decomposition*.
- 2010–2013 M.S. in Machine Automation, *Tampere University*
Thesis: *Robot Localization under Partial Observability with Weak Maps*

Honors and Fellowships

CEPS Graduate Fellowship, University of New Hampshire
Thesis and Dissertation Fellowship, Tampere University of Technology